

# Review Report

*Dilip Sahoo*

## REFERENCE

| Paper ID | Title                                                                                                                                                        | Year | Authors                                                                                                  | Journal                    |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----------------------------------------------------------------------------------------------------------|----------------------------|
| P1       | Earthquake Probability Assessment for the Indian Subcontinent Using Deep Learning                                                                            | 2020 | R. Jena, B. Pradhan, A. Al-Amri, C. W. Lee, H. Park                                                      | Sensors (MDPI)             |
| P2       | Earthquake Event Characterization and Severity Classification Using Machine Learning Algorithms: A Data-Driven Framework Based on Supervised Learning Models | 2026 | A. G. A. Diab, B. S. Tawfik, B. Hafiz                                                                    | Stat., Optim. Inf. Comput. |
| P3       | Analysis of Earthquake Forecasting in India Using Supervised Machine Learning Classifiers                                                                    | 2021 | P. Debnath, P. Chittora, T. Chakrabarti, P. Chakrabarti, Z. Leonowicz, M. Jasinski, R. Gono, E. Jasińska | Sustainability (MDPI)      |

## PROBLEM STATEMENT

**P1:** Traditional probabilistic and statistical earthquake hazard methods rely on simplifying assumptions (uniform seismicity, random occurrence in time/space) that limit their accuracy; this study develops a CNN-based probability assessment model using nine geospatial and seismic indicators to generate an updated, scalable earthquake probability map for the Indian subcontinent.

**P2:** Seismic networks detect thousands of events annually, the vast majority of which are minor and require no response; most prior ML studies suffer from data leakage (random splits, SMOTE applied before splitting) and misleading accuracy-only reporting. This study builds a chronologically-validated, imbalance-aware ML framework to automatically flag significant ( $M \geq 5.0$ ) events and classify severity in real time for early-warning triage.

**P3:** Earthquake prediction remains one of the hardest open problems in geoscience; rather than predicting exact magnitude, this study frames the problem as multi-class forecasting of magnitude range (Mild / Moderate / Fatal) across six Indian and neighbouring regions, comparing seven classical supervised classifiers to identify which is most reliable for regional deployment.

## METHODOLOGY

All three studies apply supervised learning to historical earthquake catalogs covering India and neighbouring regions, but they differ in scope and technique: P1 uses a deep convolutional neural network over nine GIS-derived thematic layers to produce a continuous probability surface; P2 applies four classical/neural classifiers with a strict chronological train/test split and SMOTE-based imbalance correction to characterize already-detected events; and P3 benchmarks seven Weka-based classifiers across six regional catalogs for coarse magnitude-range forecasting.

| Paper | Model                                                   | Core Theory                                                                                                                                                                        | Performance                                                                                        |
|-------|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| P1    | CNN (4 conv. layers, dropout + pooling, softmax output) | 9 input indicators (fault density/distance, lithology-amplification, slope, elevation, magnitude/epicenter/PGA density); binary earthquake vs. non-earthquake pixel classification | Train Acc = 96%, Test Acc = 92.05%; Precision/Recall/F1 (class 1) = 0.88/0.99/0.93                 |
| P2    | Random Forest (strongest overall)                       | 100 bootstrap-aggregated decision trees on 7 real-time network features (stations, gap, RMS, depth, etc.), class_weight=balanced + SMOTE on training partition only                | Binary: Acc = 97.39%, AUC-ROC = 0.7176, MCC = 0.1081; Multi-class: Acc = 82.51%, Macro-F1 = 0.2936 |
|       | Neural Network (MLP 50-30)                              | Feed-forward net, Adam optimizer, early stopping                                                                                                                                   | Binary Recall = 0.4211 (best minority-class detection)                                             |
|       | SVM (RBF)                                               | Maximum-margin classifier, class_weight=balanced                                                                                                                                   | Binary Recall = 0.4737 (highest raw recall, high false-alarm cost)                                 |
| P3    | Simple Logistic / LMT (best average performers)         | Logistic regression and logistic-model-tree classifiers on 22 catalog attributes (lat/long/depth/gap/rms/stations); 3-class target (Mild <4.5, Moderate 4.5–5.5, Fatal >5.5)       | Accuracy $\approx$ 98–99.9% across 5 of 6 regional datasets                                        |
|       | Random Forest                                           | Ensemble of decision trees, same feature set                                                                                                                                       | Accuracy 95.4–98.5% across regions                                                                 |
|       | ZeroR (baseline)                                        | Majority-class predictor                                                                                                                                                           | Accuracy 50.7–61.0% — confirms other models add real value                                         |

## DATASET

| Paper | Dataset Description                                                                                                                                                                                                              | Size                                                                                                                                      |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| P1    | USGS + Geological Survey of India catalog (Kolathayar et al.); 9 GIS thematic layers (fault density, lithology, slope, elevation, PGA, magnitude/epicenter density) covering the Indian subcontinent, 1900–2019 events $M > 4.5$ | 500 labeled points (350 train / 150 validation); ~20 million pixel points tested across India                                             |
| P2    | USGS Earthquake Hazards Program catalog, Northern California Seismic Network (NCSN), 1966–2007                                                                                                                                   | 18,030 events; chronological split — 12,621 train (pre-1993-09-23) / 5,409 test; only 168 events (0.93%) are significant ( $M \geq 5.0$ ) |
| P3    | USGS earthquake feed, 1900–2020, six regional subsets: Andaman & Nicobar, Gujarat, North India, North-East India, Nepal/UP/Bihar, and Upper Half of India; 22 attributes per record                                              | 185 – 14,085 instances per region (total across 6 datasets $\approx 26,978$ records)                                                      |

## RESULTS

| Paper | Best Model            | Key Metric                                                                                                                 | Note                                                                                                                                              |
|-------|-----------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| P1    | CNN classifier        | Test Acc = 92.05%; Very-high probability zone = 712,375 km <sup>2</sup> (19.8%); High = 591,240.5 km <sup>2</sup> (16.43%) | 92% of predicted earthquakes fall in Himalayan high/very-high zones; validated against GSI zoning map and 5 domain experts (80% highly satisfied) |
| P2    | Random Forest         | Binary Acc = 97.39%, AUC-ROC = 0.7176; Multi-class Acc = 82.51%, Macro-F1 = 0.2936                                         | MCC (0.1081) far below accuracy — reveals RF barely beats a naive majority baseline (~99.3% acc.) on the true positive minority class             |
| P3    | Simple Logistic / LMT | Up to 99.94% accuracy (Andaman & Nicobar); minimum 96.36% (Gujarat)                                                        | Performance varies by region/dataset size; Bayes Net close second; no single classifier universally best across all six regions                   |

Across the three papers, ensemble and tree-based methods (Random Forest in P2, LMT/Random Forest in P3) and deep spatial models (CNN in P1) consistently outperform simple linear or distance-based baselines. However, P2's rigorous chronological validation exposes a gap that P1 and P3 do not directly address: high raw accuracy can mask poor detection of the rare, operationally critical significant-earthquake class, underscoring the importance of imbalance-aware metrics (F1, MCC, AUC-ROC) alongside accuracy.

## LIMITATIONS

| Paper | Key Limitations                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P1    | Time- and data-intensive CNN training; limited to nine indicators derived largely from digitized/interpolated GIS layers; probability estimation only (not time-of-occurrence prediction); assumptions in thematic-layer preparation can introduce noise.                                                                                                                                                        |
| P2    | Task is event characterization (post-detection), not occurrence forecasting; trained only on the Northern California network — a sparsification test showed AUC-ROC degrading from 0.7176 to 0.648 at ~75% station density; Moderate/Strong test classes (n = 32, n = 6) are statistically too small for reliable per-class metrics; classifier probabilities are poorly calibrated without post-hoc correction. |
| P3    | Coarse 3-class magnitude-range target rather than continuous/precise magnitude; uses classical Weka classifiers only (no deep learning); six regional datasets vary greatly in size (185–14,085), making cross-region comparison uneven; ZeroR baseline shows some 'high accuracy' results are inflated by class imbalance within a given region.                                                                |

## FUTURE SCOPE

**P1:** Incorporate additional geophysical/geological indicators (soil composition, aftershock behaviour), build a national active-fault database, and extend the CNN framework with real-time monitoring and transfer learning to other seismic regions.

**P2:** Enrich features with geophysical variables (b-values, strain rates, fault proximity), engineer temporal/sliding-window seismicity features, explore deep learning on raw waveforms, validate across multiple seismic networks/regions, and apply probability calibration for decision-support deployment.

**P3:** Extend beyond classical Weka classifiers to deep learning and larger, more balanced regional datasets; refine class boundaries for magnitude range; explore ensemble/hybrid approaches (as seen in P1 and P2) to reduce the region-to-region performance variability observed in this study.